ACADEMIC CAPSTONE PROJECT PRESENTATION

SMARTBIN

Smart Liquidation Bin Store E-Commerce Management System

Featuring Automated Dynamic Daily Price Drop Architecture, In-Store Bin Mapping, and Reverse Logistics Management.

- Project Core Innovation: Automated Descending Daily Pricing Engine (Friday to Thursday Price Drops)
- **Technology Stack: Node.js, Express.js, MySQL 8.0, HTML5, CSS3, JavaScript (ES6+), Bootstrap 5, Razorpay**
- **Target Roles: Customers (Online Shoppers), Store Managers (Daily Pricing & Stock), Super Administrators**

Project Overview & Background

- **The Liquidation Bin Retail Model:** Liquidation stores purchase truckloads of customer returns, overstocks, and clearance merchandise from major retailers (Amazon, Walmart, Target) and resell them at deep discounts.
- **The Core Concept of Descending Daily Pricing:** Unlike regular e-commerce with static prices, bin stores sell items on a weekly descending price cycle where every single product drops in price each day until the bins are completely cleared.
- **The Weekly Clearance Cycle:** New inventory arrives every Friday at the highest price. Prices systematically drop every day (Saturday, Sunday, Monday, Tuesday, Wednesday) until Thursday when remaining items are cleared at clearance rates.
- **The Reverse Logistics Bottleneck:** Traditional stores manage these price changes manually with stickers or announcements, causing billing confusion, crowd stampedes, and inventory miscalculations.
- **SmartBin's Digital Solution:** A centralized web platform that automates daily price recalculations in MySQL, displays live daily prices to online customers, maps physical bin locations, and enables 24/7 shopping.

Project Objectives & Scope

- **Automate Dynamic Daily Price Drop Schedules:** Develop an automated pricing engine that checks the current calendar day and updates selling prices across products at midnight without human labor or manual price tagging.
- **Support Product-Specific Daily Price Overrides:** Enable Store Managers to define tailored daily price schedules per product for high-value merchandise (e.g., electronics) using the daily_prices database table.
- **Provide Real-Time Inventory & Bin Transparency:** Display exact physical bin locations (Bin A, Bin B, Bin C) and condition ratings (Brand New, Open Box, Refurbished) so buyers know where items are located and their true condition.
- **Secure Multi-Role Governance:** Provide isolated portals for Customers, Store Managers, and Super Administrators protected by JWT authentication and salted bcrypt password hashing.
- **Frictionless E-Commerce & Customer Retention:** Provide dynamic cart calculation, promotional coupon discounts, Razorpay online payments, and automated Lucky Draw raffle competitions.

Existing System vs. Proposed System

Limitations of Existing Traditional Bin Stores:

– Manual Price Re-Labeling: Workers manually re-tag hundreds of items every evening or shout out new daily rates, leading to severe labor overhead and pricing errors.

– Zero In-Store Transparency: Shoppers have no way of knowing what products will be priced at on subsequent days, making planning impossible.

– Blind In-Person Shopping: Customers must travel physically to stores and rummage through unorganized wooden bins without knowing available stock.

– Billing Discrepancies: Cashiers manually enter daily prices at registers, frequently miscalculating totals during crowded morning rushes.

Innovations of Proposed SmartBin Platform:

✓ Automated Midnight Price Engine: The database scheduler recalculates and applies today's liquidated price automatically at 12:00 AM.

✓ Transparent 7-Day Price Schedule: Customers view the exact price for today and upcoming days of the week on the storefront before purchasing.

✓ 24/7 Digital Catalog & Bin Mapping: Customers browse items by physical bin number (Bin A, B), condition grade, and reserve or buy online.

✓ Instant Online Checkout: Razorpay payment gateway automatically applies today's liquidated deal price, generating digital invoices.

Main Feature: Dynamic Daily Pricing Architecture

1. Exact Weekly Liquidation Price Drop Schedule (Standard Bins):

- **Friday (Restock Day):** ₹999 — Fresh inventory loaded into bins from retail pallets; highest value day.
- **Saturday (Weekend Rush):** ₹799 — High-demand weekend shopping price drop.
- **Sunday (Weekend Drop):** ₹599 — Sunday price cut to accelerate weekend volume.
- **Monday (Markdown Monday):** ₹399 — Weekday price reduction to drive steady traffic.
- **Tuesday (Value Tuesday):** ₹299 — Budget clearance pricing tier.
- **Wednesday (Blowout Day):** ₹199 — Deep discount blowout day to clear bins.
- **Thursday (Clearance Day):** ₹99 — Final liquidation price before Friday's new inventory restock.

2. Product-Specific Daily Price Overrides (daily_prices Table):

- **High-Value Product Overrides:** For electronics or premium products, Store Managers configure custom daily prices per day in daily_prices (e.g., Fri: ₹4,999 ➔ Sat: ₹3,999 ➔ Sun: ₹2,999 ➔ Mon: ₹1,999 ➔ Tue: ₹1,499 ➔ Wed: ₹999 ➔ Thu: ₹499).
- **Automated Daily Application API (/api/daily-prices/apply-today):** Evaluates current day name (Sunday to Saturday), executes UPDATE products SET price = ? WHERE product_id = ?, and instantly syncs live catalog prices.

System Architecture & Execution Workflow

- **Phase 1: Ingestion & Daily Price Setup:** Store Managers catalog incoming liquidation pallets, assign physical Bin Numbers (Bin A, Bin B), condition grades, and configure the product's daily price schedule from Friday to Thursday.
- **Phase 2: Automated Midnight Price Recalculation:** At 12:00 AM, the Express server checks the calendar day against the daily price schedule. Product prices in the database are updated automatically to reflect today's liquidated rate.
- **Phase 3: Customer Discovery & Cart Management:** Shoppers browse the storefront showing today's deal price, filter by bin or condition, view upcoming daily price drops, add items to cart, and apply discount promo codes.
- **Phase 4: Payment Gateway & Order Synchronization:** Transactions are completed via Razorpay API. In an atomic transaction, the system creates order/payment records, charges today's dynamic price, and decrements stock.
- **Phase 5: Fulfillment & Lucky Draw Participation:** Store Managers review orders on the dispatch dashboard and update statuses (Pending ➔ Confirmed ➔ Shipped ➔ Delivered). Customer orders automatically enter weekly Lucky Draws.

Module Description & Role Responsibilities

- **Customer Portal (Storefront):**
  - Today's Deal Display: Prominently view today's active liquidation price banner and 7-day schedule.
  - Catalog Browsing & Filters: Filter items by Bin Number (Bin A, B, C), Category, Brand, and Condition Grade.
  - Cart, Wishlist & Vouchers: Subtotal calculation using today's price, stock checks, and coupon discounts.
  - Order Tracking & Raffles: Self-service order status tracking and Lucky Draw winner boards.
- **Store Manager Portal:**
  - Daily Price Management: Configure custom Monday–Sunday daily prices for individual products.
  - Apply Today's Prices: One-click button to trigger /api/daily-prices/apply-today to synchronize product prices.
  - Product Management & Bins: Add, edit, restock products, and assign physical store bin locations.
  - Order Fulfillment: Review customer delivery details and update order dispatch stages.
- **Super Administrator Dashboard:**
  - Platform Analytics: Monitor total revenue, daily orders, active managers, and registered customers.
  - Global Price Schedule Master: Configure default weekly price drop curves (Friday to Thursday).
  - Manager Oversight: Onboard, update credentials, and manage privileges of store managers.
  - Marketing & Raffles: Launch coupon campaigns and trigger automated lucky draw winner selection.

Database Design & Daily Pricing Schema

13 Normalized Relational Tables in MySQL (liquidation_bin):

- **daily_prices Entity (Core Feature Table):** Stores price_id (PK), product_id (FK), day_name (ENUM: 'Monday'...'Sunday'), price (DECIMAL 10,2), and updated_at timestamp. Enables per-product daily price overrides.
- **products Entity:** Stores product_id (PK), manager_id (FK), product_name, category, brand, price (current liquidated price), quantity, image, and bin_number.
- **Authentication & Roles Entities:** admins, managers, and users tables store user credentials with 60-character bcrypt salted password hashing and unique email indexes.
- **Commerce & Transactional Entities:** cart, cart_items, wishlist, orders, order_items, and payments tables maintain complete transactional integrity with cascading foreign keys.
- **Marketing & Gamification Entities:** offers, promotional_offers, lucky_draws, and winners tables manage coupon codes, raffle contests, and winner history.
- **Database Normalization & Integrity:** Structured in Third Normal Form (3NF) with foreign key constraints (ON DELETE CASCADE / SET NULL) to ensure zero data redundancy and ACID reliability.

Technology Stack & Development Tools

- **Frontend Layer:** HTML5 for semantic layouts, CSS3 for custom responsive styling, Bootstrap 5 for grid systems and modal interfaces, Vanilla JavaScript (ES6+) for dynamic DOM rendering and Fetch API, and FontAwesome 6 icons.
- **Backend & API Layer:** Node.js as the non-blocking asynchronous event loop runtime; Express.js as the RESTful API framework; Express Router for modular daily-price and product routes; Multer for product photo uploads; CORS and Body-Parser.
- **Database & Security:** MySQL 8.0 ACID-compliant relational database engine; mysql2/promise driver supporting connection pooling; JSON Web Tokens (JWT) for stateless authentication; bcrypt.js for 10-round password hashing; Dotenv for environment protection.
- **Payment Gateway & Tools:** Razorpay Payment Gateway API for secure online card/UPI checkout; Postman for REST endpoint testing; Visual Studio Code as the primary IDE; Git and GitHub for version control; and Nodemon for hot-reloading development.

Future Enhancements & Project Conclusion

Future Enhancements & Scalability Roadmap:

- Automated Midnight Cron Jobs: Configure node-cron to execute /api/daily-prices/apply-today at precisely 12:00 AM daily without requiring manager triggers.
- AI Dynamic Pricing Optimization: Deploy machine learning models to analyze remaining stock velocity and automatically recommend optimal daily price drop rates.
- Cross-Platform Mobile App: Build Flutter / React Native apps with automated push notifications notifying users at midnight of new daily price drops.
- In-Store Omnichannel QR Code Scanners: Place QR tags on physical bins so in-store walk-in customers can scan items to check upcoming price drop schedules.

Project Conclusion & Takeaways:

✓ Successfully digitized reverse logistics liquidation retail with an automated dynamic daily pricing engine.

✓ Demonstrated full-stack engineering with a responsive frontend, RESTful Node.js backend, and a 13-table 3NF MySQL database.

✓ Delivered real business impact: descending daily price drops, physical bin mapping, online payments, and gamification.

Thank You! — Questions & Answers Session Open for Discussion